

# What else can the lopsided technique speed up? Survey report (2026-10-06)

Question and method in the **protocol** (README.md). Round 1 (interface + four IC insertion points, all proposals, merged PR #1485) is the companion study. Statement status is labelled **[proposed]** / **[derived]** / **[verified]** / **[measured]** (none in this round).

## Bottom line

Twelve candidate transfers were screened. **Zero repository workloads pass all three gates; the reusable gate itself is the positive deliverable.** Five fail the shape gate, two fail the ring gate, none from the repo reaches the size gate with a valid shape, and five are out of scope. The one passing shape is synthetic, proving the gate is not vacuous **[verified]** by unit test.

## Screening funnel

**Screening funnel -- 12 candidate transfers, zero repo passers, gate named per reject**

Gate order matches screen\_workload: shape, then ring, then size. Numbered candidates refer to the REPORT table.

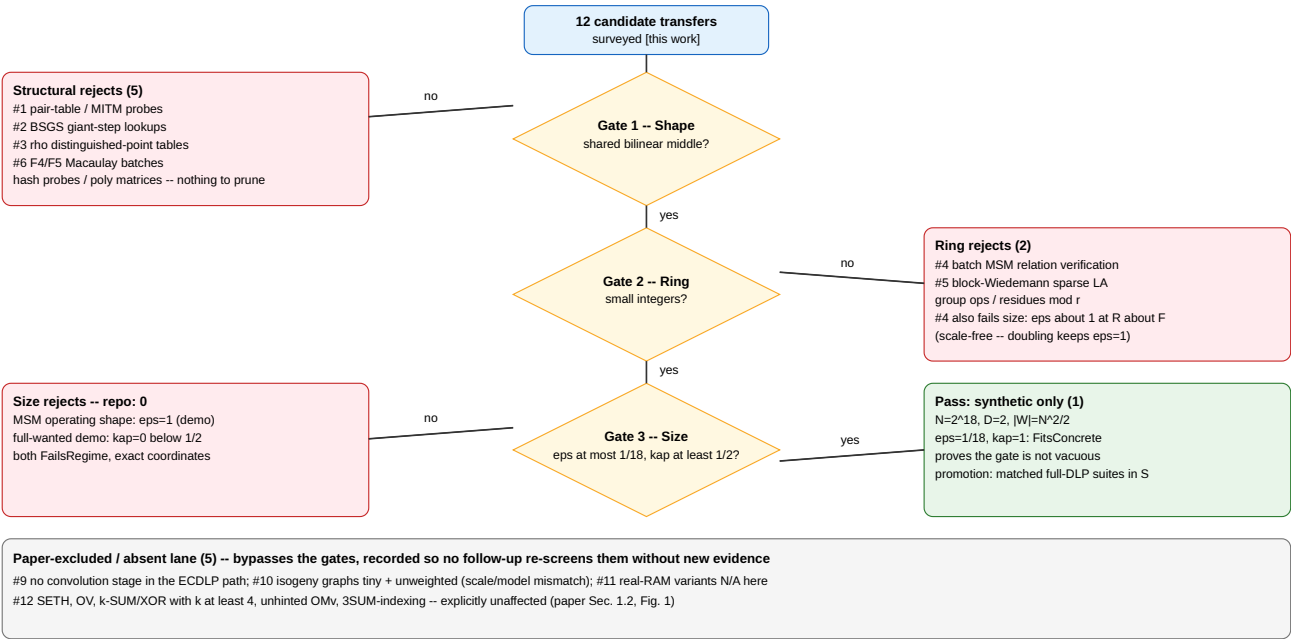


Figure 1 -- the funnel. Gate order matches screen\_workload: shape, then ring, then size. Numbered candidates refer to the table below. Editable source: figure.mmd.

## Ranked table

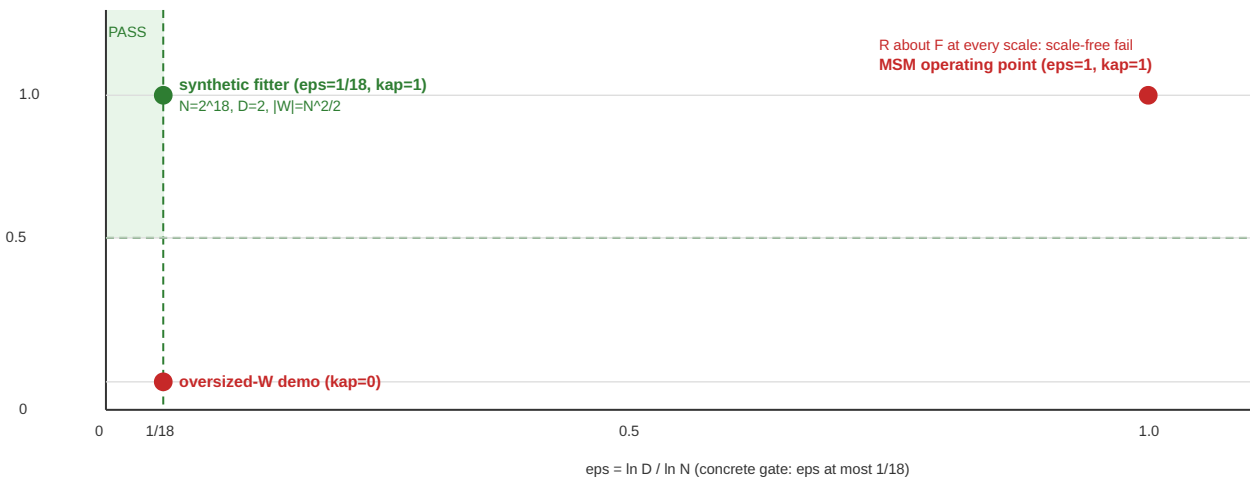
| #  | Candidate (paper source to repo target)                                                                        | Verdict [derived] | Failing gate + reason                                                                                                                                                       |
|----|----------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Hinted-OMv workflow to online pair-table probes (orbit_pair_table m=2/m=3; native_signed_mitm batched lookups) | Negative          | <b>Shape:</b> frozen table + online queries matches the workflow, but each probe is a hash lookup over an unstructured key space, not an inner product over a shared middle |
| 2  | BSGS giant-step lookups vs baby table                                                                          | Negative          | <b>Shape:</b> hash probes; the table is large ( $\sqrt{n}$ ), the opposite of thin                                                                                          |
| 3  | Rho distinguished-point / collision tables                                                                     | Negative          | <b>Shape:</b> hash probes on walk states                                                                                                                                    |
| 4  | Thin batch verification to batch MSM relation checks                                                           | Negative          | <b>Ring</b> (then size): group operations, not small-integer products; at R about F, eps about 1 scale-free                                                                 |
| 5  | Sparse-LA block Wiedemann (round-1 re-screen)                                                                  | Negative          | <b>Ring:</b> large sparse matvecs over $\mathbb{Z}/r\mathbb{Z}$                                                                                                             |
| 6  | Rectangular MM to F4/F5 Macaulay batch reduction                                                               | Negative          | <b>Shape/Ring:</b> polynomial matrices over fields; no shared thin integer middle                                                                                           |
| 7  | 3SUM $O(n^{1.9992})$ to Semaev/MITM point decomposition                                                        | Negative          | <b>Reduction missing:</b> integers vs curve points with algebraic constraints                                                                                               |
| 8  | 3XOR speedup to Boolean SAT/F4 systems                                                                         | Negative          | <b>Reduction missing:</b> 3XOR instances are not our polynomial systems                                                                                                     |
| 9  | MonoConvolution $O(n^{1.5-\text{eps}})$ to convolution stages                                                  | Out of scope      | No convolution stage in the ECDLP path                                                                                                                                      |
| 10 | (min,+)/APSP to isogeny-route search                                                                           | Negative          | <b>Scale + model:</b> tiny unweighted graphs vs large-n polynomial weights                                                                                                  |
| 11 | Real-valued Las Vegas variants (Sec. 5.2)                                                                      | Out of scope      | Finite-field repository; real-RAM model absent                                                                                                                              |
| 12 | SETH, OV, k-SUM/XOR k at least 4, unhinted OMv, 3SUM-indexing                                                  | Out of scope      | Explicitly unaffected per the paper                                                                                                                                         |

## Screening map

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### Screening map -- gate evaluations in the (eps, kap) plane, NOT measurements

Points are exact closed-form gate evaluations of workload shapes. Structural rejects have no coordinates.



**No coordinates -- structural rejects (hash probes / poly matrices, NoSharedMiddle):**

#1 pair-table + MITM probes, #2 BSGS lookups, #3 rho tables, #6 Macaulay batches. Ring rejects (#4 MSM, #5 Wiedemann) fail before size.

Figure 2 -- the (eps, kap) plane. Green: pass region. Points are exact gate evaluations of workload shapes (not measurements). Structural rejects have no coordinates and are listed, not plotted.

## Promotion protocol for any future passer [proposed]

A candidate returning `FitsConcrete` earns a follow-up measurement protocol, not a claim: freeze curve/subgroup/base/targets/seeds, run matched baseline/candidate full-DLP suites with every phase priced in S against the generic floor and matched rho, classify per AGENTS.md Sec. 3, and update scoreboard/leaderboard/browser in that PR.

## Boundary, table, ratio (AGENTS.md Secs. 1-8)

- Boundaries stated before measuring: generic floor + matched rho, unchanged; unit S.
- No table row produced: gate evaluations are not verified DLP answers.
- Falsification target: a repo workload returning `FitsConcrete` with a concrete (N, D, W) mapping would overturn this round's headline and open the promotion protocol.
- All phases unpriced (null); wall time appears nowhere.

## Graphs checked -- no change [verified by inspection]

- docs/index-calculus-scoreboard.html + docs/ic/progress-timeline.json: no new ratio.
- docs/ic-leaderboard.html / LEADERBOARD.md / leaderboard.json: no new measurement.
- docs/browser/data.json: no new curve, session, round, or IC1 identity.
- docs/curves/registry.json: no curve named; family notation only.
- Figures 1-2 are new visuals for this search round, not canonical-graph updates.

## Open checks [proposed]

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1. Future batch workloads with  $N$  small-integer inner products over one shared  $D$ -dimensional middle go through `screen_workload` first; call and verdict belong in their protocol.
2. Candidate #4's eps about 1 is scale-free only at  $R$  about  $F$ ; a workload with  $R$  much larger than  $F^2$  would move eps -- none exists in the pipeline today.
3. The paper's general regime (eps below 0.1204, size-dependent  $\gamma$ ) is not encoded in the gate; a candidate failing the concrete 1/18 line but plausibly inside the general regime needs a Section-4 analysis first.

## Sources

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- Paper: <https://arxiv.org/pdf/2610.06783> (v1), Fig. 1 (p. 9), Secs. 5.1-5.4.
- Round 1: `research/lopsided_thin_product_20261006/` (merged PR #1485).
- Code: `screen_workload` in `src/cryptanalysis/lopsided_thin_product.rs` (this PR).